

1 First Order Logic

- (A1) $A \rightarrow (B \rightarrow A)$
 (A2) $(A \rightarrow (B \rightarrow C)) \rightarrow ((A \rightarrow B) \rightarrow (A \rightarrow C))$
 (A3) $(\neg B \rightarrow \neg A) \rightarrow (A \rightarrow B)$

Claim. $\vdash P \rightarrow P$.

Proof.

- | | | |
|-----|---|--|
| (1) | $P \rightarrow (P \rightarrow P)$ | instance of (A1) with $A := P$, $B := P$ |
| (2) | $P \rightarrow ((P \rightarrow P) \rightarrow P)$ | instance of (A1) with $A := P$, $B := (P \rightarrow P)$ |
| (3) | $(P \rightarrow ((P \rightarrow P) \rightarrow P)) \rightarrow ((P \rightarrow (P \rightarrow P)) \rightarrow (P \rightarrow P))$ | instance of (A2) with $A := P$, $B := (P \rightarrow P)$, $C := P$ |
| (4) | $(P \rightarrow (P \rightarrow P)) \rightarrow (P \rightarrow P)$ | from (2) and (3) by Modus Ponens |
| (5) | $P \rightarrow P$ | from (1) and (4) by Modus Ponens |

Hence $\vdash P \rightarrow P$. □

Claim. $\{P \rightarrow Q, Q \rightarrow R\} \vdash P \rightarrow R$.

Proof.

- | | | |
|-----|---|---|
| (1) | $(Q \rightarrow R) \rightarrow (P \rightarrow (Q \rightarrow R))$ | instance of (A1) with $A := (Q \rightarrow R)$, $B := P$ |
| (2) | $Q \rightarrow R$ | given |
| (3) | $P \rightarrow (Q \rightarrow R)$ | from (1) and (2) by Modus Ponens |
| (4) | $(P \rightarrow (Q \rightarrow R)) \rightarrow ((P \rightarrow Q) \rightarrow (P \rightarrow R))$ | instance of (A2) with $A := P$, $B := Q$, $C := R$ |
| (5) | $(P \rightarrow Q) \rightarrow (P \rightarrow R)$ | from (3) and (4) by Modus Ponens |
| (6) | $P \rightarrow Q$ | given |
| (7) | $P \rightarrow R$ | from (5) and (6) by Modus Ponens |

Thus $P \rightarrow R$ is derivable from the assumptions. □

Claim. The formulas underneath are logically equivalent

$$\forall x \forall y (q(x, y) \rightarrow p(x)) \quad \text{and} \quad \forall x (\exists y q(x, y) \rightarrow p(x))$$

Proof.

$$\begin{aligned} \forall x \forall y (q(x, y) \rightarrow p(x)) &\equiv \forall x \forall y (\neg q(x, y) \vee p(x)) \\ &\equiv \forall x (\forall y (\neg q(x, y) \vee p(x))) \\ &\equiv \forall x (\neg \exists y q(x, y) \vee p(x)) \\ &\equiv \forall x (p(x) \vee \neg \exists y q(x, y)) \\ &\equiv \forall x (\exists y q(x, y) \rightarrow p(x)). \end{aligned}$$

Thus the two statements are equivalent. □

Lemma 1. $\mathcal{I}, \sigma \models \forall x B$ iff for all $d \in D$, $\mathcal{I}, \sigma[x \leftarrow d] \models B$.

Lemma 2. $\mathcal{I}, \sigma \models \exists x B$ iff for some $d \in D$, $\mathcal{I}, \sigma[x \leftarrow d] \models B$.

Claim: $\forall x (A(x) \wedge B(x)) \equiv (\forall x A(x)) \wedge (\forall x B(x))$.

Proof.

We show both directions of logical implication using the semantic definition of $\models$.

(1) $\forall x (A(x) \wedge B(x)) \Rightarrow (\forall x A(x)) \wedge (\forall x B(x))$ Let $\mathcal{I}$ be any interpretation with domain $D \neq \emptyset$, and let σ be any variable assignment. Assume

$$\mathcal{I}, \sigma \models \forall x (A(x) \wedge B(x)).$$

By the semantics of the universal quantifier, this means:

$$\text{for all } d \in D, \quad \mathcal{I}, \sigma[x \mapsto d] \models A(x) \wedge B(x).$$

By the semantics of conjunction:

$$\text{for all } d \in D, \quad \mathcal{I}, \sigma[x \mapsto d] \models A(x) \quad \text{and} \quad \mathcal{I}, \sigma[x \mapsto d] \models B(x).$$

Hence:

$$\mathcal{I}, \sigma \models \forall x A(x) \quad \text{and} \quad \mathcal{I}, \sigma \models \forall x B(x).$$

Thus, by semantics of $\wedge$,

$$\mathcal{I}, \sigma \models (\forall x A(x)) \wedge (\forall x B(x)).$$

Since $\mathcal{I}$ and σ were arbitrary, we obtain:

$$\models \forall x (A(x) \wedge B(x)) \rightarrow ((\forall x A(x)) \wedge (\forall x B(x))).$$

(2) $(\forall x A(x)) \wedge (\forall x B(x)) \Rightarrow \forall x (A(x) \wedge B(x))$ Again let $\mathcal{I}$ be any interpretation with domain $D \neq \emptyset$ and σ any assignment. Assume

$$\mathcal{I}, \sigma \models (\forall x A(x)) \wedge (\forall x B(x)).$$

By semantics of $\wedge$:

$$\mathcal{I}, \sigma \models \forall x A(x) \quad \text{and} \quad \mathcal{I}, \sigma \models \forall x B(x).$$

Thus, by semantics of $\forall$:

$$\text{for all } d \in D, \quad \mathcal{I}, \sigma[x \mapsto d] \models A(x) \quad \text{and} \quad \mathcal{I}, \sigma[x \mapsto d] \models B(x).$$

Therefore, for all $d \in D$,

$$\mathcal{I}, \sigma[x \mapsto d] \models A(x) \wedge B(x).$$

Applying the semantics of $\forall$ again yields:

$$\mathcal{I}, \sigma \models \forall x (A(x) \wedge B(x)).$$

Thus,

$$\models ((\forall x A(x)) \wedge (\forall x B(x))) \rightarrow \forall x (A(x) \wedge B(x)).$$

Claim: $\exists x (A(x) \vee B(x)) \equiv (\exists x A(x)) \vee (\exists x B(x))$.

Proof.

We prove both directions using the semantic definition of $\models$.

(1) $\exists x(A(x) \vee B(x)) \Rightarrow (\exists x A(x)) \vee (\exists x B(x))$ Let $\mathcal{I}$ be any interpretation with domain $D \neq \emptyset$, and let σ be any variable assignment. Assume

$$\mathcal{I}, \sigma \models \exists x (A(x) \vee B(x)).$$

By the semantics of the existential quantifier, there exists some $d \in D$ such that

$$\mathcal{I}, \sigma[x \mapsto d] \models A(x) \vee B(x).$$

By the semantics of disjunction, this means:

$$\mathcal{I}, \sigma[x \mapsto d] \models A(x) \quad \text{or} \quad \mathcal{I}, \sigma[x \mapsto d] \models B(x).$$

- If $\mathcal{I}, \sigma[x \mapsto d] \models A(x)$, then by the semantics of $\exists$ we have $\mathcal{I}, \sigma \models \exists x A(x)$.
- If $\mathcal{I}, \sigma[x \mapsto d] \models B(x)$, then $\mathcal{I}, \sigma \models \exists x B(x)$.

In either case,

$$\mathcal{I}, \sigma \models (\exists x A(x)) \vee (\exists x B(x)).$$

Since $\mathcal{I}$ and σ were arbitrary, we obtain

$$\models \exists x(A(x) \vee B(x)) \rightarrow ((\exists x A(x)) \vee (\exists x B(x))).$$

(2) $(\exists x A(x)) \vee (\exists x B(x)) \Rightarrow \exists x(A(x) \vee B(x))$ Again let $\mathcal{I}$ be any interpretation with domain $D \neq \emptyset$ and σ any assignment. Assume

$$\mathcal{I}, \sigma \models (\exists x A(x)) \vee (\exists x B(x)).$$

By the semantics of $\vee$,

$$\mathcal{I}, \sigma \models \exists x A(x) \quad \text{or} \quad \mathcal{I}, \sigma \models \exists x B(x).$$

- Suppose $\mathcal{I}, \sigma \models \exists x A(x)$. Then there exists $d \in D$ such that $\mathcal{I}, \sigma[x \mapsto d] \models A(x)$. Hence

$$\mathcal{I}, \sigma[x \mapsto d] \models A(x) \vee B(x),$$

so by the semantics of $\exists$,

$$\mathcal{I}, \sigma \models \exists x(A(x) \vee B(x)).$$

- Suppose instead $\mathcal{I}, \sigma \models \exists x B(x)$. Then there exists $d \in D$ such that $\mathcal{I}, \sigma[x \mapsto d] \models B(x)$, and hence again

$$\mathcal{I}, \sigma[x \mapsto d] \models A(x) \vee B(x),$$

yielding

$$\mathcal{I}, \sigma \models \exists x(A(x) \vee B(x)).$$

Thus in all cases,

$$\mathcal{I}, \sigma \models \exists x(A(x) \vee B(x)).$$

Therefore,

$$\models ((\exists x A(x)) \vee (\exists x B(x))) \rightarrow \exists x(A(x) \vee B(x)).$$

Proposition 1. In the structure $\mathcal{I}$ and assignment σ (in class example),

$$\mathcal{I}, \sigma \models \forall x \exists y \text{lt}(x, y) \quad \text{and} \quad \mathcal{I}, \sigma \not\models \exists y \forall x \text{lt}(x, y).$$

Proof. First formula. By the semantics of $\forall$,

$$\mathcal{I}, \sigma \models \forall x \exists y \text{lt}(x, y)$$

iff for every $d \in D$,

$$\mathcal{I}, \sigma[x \leftarrow d] \models \exists y \text{lt}(x, y).$$

Fix $d \in D$. By the semantics of $\exists$ this is equivalent to the existence of some $e \in D$ such that

$$\mathcal{I}, \sigma[x \leftarrow d][y \leftarrow e] \models \text{lt}(x, y).$$

Choose $e = d$. Then $d \leq d$ in $\mathbb{N}$, so $\text{lt}^{\mathcal{I}}(d, d)$ holds, hence $\mathcal{I}, \sigma[x \leftarrow d][y \leftarrow d] \models \text{lt}(x, y)$. Thus for every d such an e exists, and therefore $\mathcal{I}, \sigma \models \forall x \exists y \text{lt}(x, y)$.

Second formula. Suppose, for contradiction, that

$$\mathcal{I}, \sigma \models \exists y \forall x \text{lt}(x, y).$$

By the semantics of $\exists$, there is some $d \in D$ such that

$$\mathcal{I}, \sigma[y \leftarrow d] \models \forall x \text{lt}(x, y).$$

By the semantics of $\forall$, this means that for every $e \in D$,

$$\mathcal{I}, \sigma[y \leftarrow d][x \leftarrow e] \models \text{lt}(x, y),$$

i.e. $\text{lt}^{\mathcal{I}}(e, d)$ holds for all $e \in D$.

Take $e = d + 1 \in \mathbb{N}$. Then $\text{lt}^{\mathcal{I}}(d + 1, d)$ would have to hold, i.e. we would need $d + 1 \leq d$, which is false in $\mathbb{N}$. Hence

$$\mathcal{I}, \sigma[y \leftarrow d][x \leftarrow d + 1] \not\models \text{lt}(x, y),$$

contradicting the assumption that $\mathcal{I}, \sigma[y \leftarrow d] \models \forall x \text{lt}(x, y)$.

Therefore no such d exists, so

$$\mathcal{I}, \sigma \not\models \exists y \forall x \text{lt}(x, y).$$

□

Proposition 2. Let Γ be a set of first-order formulas and A a formula. Then

$$\Gamma \models A \iff \Gamma \cup \{\neg A\} \text{ is unsatisfiable.}$$

Proof. Recall:

- $\mathcal{I}, \sigma \models \Gamma$ means $\mathcal{I}, \sigma \models B$ for every $B \in \Gamma$.
- $\Gamma \models A$ means: for all $\mathcal{I}, \sigma$, if $\mathcal{I}, \sigma \models \Gamma$ then $\mathcal{I}, \sigma \models A$.
- Γ is *unsatisfiable* means: there are no $\mathcal{I}, \sigma$ with $\mathcal{I}, \sigma \models \Gamma$.

($\Rightarrow$) Assume $\Gamma \models A$. Suppose, for contradiction, that $\Gamma \cup \{\neg A\}$ is satisfiable. Then there exist $\mathcal{I}, \sigma$ such that $\mathcal{I}, \sigma \models \Gamma$ and $\mathcal{I}, \sigma \models \neg A$. From $\Gamma \models A$ and $\mathcal{I}, \sigma \models \Gamma$ we obtain $\mathcal{I}, \sigma \models A$. But $\mathcal{I}, \sigma \models \neg A$ iff $\mathcal{I}, \sigma \not\models A$, a contradiction. Hence no such $\mathcal{I}, \sigma$ exist, so $\Gamma \cup \{\neg A\}$ is unsatisfiable.

($\Leftarrow$) Assume $\Gamma \cup \{\neg A\}$ is unsatisfiable. To show $\Gamma \models A$, let $\mathcal{I}, \sigma$ be arbitrary such that $\mathcal{I}, \sigma \models \Gamma$. If $\mathcal{I}, \sigma \not\models A$, then by the negation clause we would have $\mathcal{I}, \sigma \models \neg A$, and hence

$$\mathcal{I}, \sigma \models \Gamma \cup \{\neg A\},$$

contradicting the unsatisfiability of $\Gamma \cup \{\neg A\}$. Therefore $\mathcal{I}, \sigma \models A$. Since $\mathcal{I}, \sigma$ was arbitrary, we conclude $\Gamma \models A$.

Thus $\Gamma \models A$ iff $\Gamma \cup \{\neg A\}$ is unsatisfiable. □

Let

$$A := \exists x \forall y p(x, f(y)), \quad B := \forall y p(c, f(y)),$$

where c is a fresh constant symbol not appearing in A . We prove that A and B are *equi-satisfiable*.

Theorem 1. *A is satisfiable iff B is satisfiable.*

Proof. (**A satisfiable $\Rightarrow B$ satisfiable**). Assume A is satisfiable. Thus there exists a structure $\mathcal{I}$ and an assignment σ such that

$$\mathcal{I}, \sigma \models \exists x \forall y p(x, f(y)).$$

By the semantics of $\exists$, there exists an element $d \in D^{\mathcal{I}}$ such that

$$\mathcal{I}, \sigma[x := d] \models \forall y p(x, f(y)).$$

Thus, for every $e \in D^{\mathcal{I}}$,

$$\mathcal{I}, \sigma[x := d, y := e] \models p(x, f(y)).$$

Now define a new structure $\mathcal{I}'$ obtained from $\mathcal{I}$ by interpreting the new constant c as d , i.e. $c^{\mathcal{I}'} = d$, and leaving all other interpretations unchanged. Then for every $e \in D^{\mathcal{I}'}$ we have

$$\mathcal{I}', \sigma[y := e] \models p(c, f(y)).$$

Hence

$$\mathcal{I}' \models \forall y p(c, f(y)),$$

so $\mathcal{I}' \models B$, and therefore B is satisfiable.

(**B satisfiable $\Rightarrow A$ satisfiable**). Assume that B is satisfiable. Thus there exists a structure $\mathcal{J}$ such that

$$\mathcal{J} \models \forall y p(c, f(y)).$$

Let $d := c^{\mathcal{J}}$ be the interpretation of c in $\mathcal{J}$. Then for every $e \in D^{\mathcal{J}}$,

$$\mathcal{J}, \sigma[y := e] \models p(c, f(y)).$$

Since $c^{\mathcal{J}} = d$, this means

$$\mathcal{J}, \sigma[x := d, y := e] \models p(x, f(y)).$$

Therefore,

$$\mathcal{J}, \sigma[x := d] \models \forall y p(x, f(y))$$

for every assignment σ . By choosing the witness $x := d$ we obtain

$$\mathcal{J}, \sigma \models \exists x \forall y p(x, f(y)).$$

Thus $\mathcal{J}, \sigma \models A$, and hence A is satisfiable.

Since each direction holds, A and B are equi-satisfiable. □

Let

$$C := \forall y \exists x p(x, y), \quad D := \forall y p(g(y), y),$$

where g is a fresh unary function symbol not appearing in C . We prove that C and D are *equi-satisfiable*.

Theorem 2. *C is satisfiable iff D is satisfiable.*

Proof. (C satisfiable $\Rightarrow D$ satisfiable). Assume that C is satisfiable. Then there exists a structure $\mathcal{I}$ and an assignment σ such that

$$\mathcal{I}, \sigma \models \forall y \exists x p(x, y).$$

By the semantics of $\forall$, this means that for every $d \in D^{\mathcal{I}}$,

$$\mathcal{I}, \sigma[y := d] \models \exists x p(x, y).$$

By the semantics of $\exists$, for each $d \in D^{\mathcal{I}}$ there exists some element $e \in D^{\mathcal{I}}$ (which may depend on d) such that

$$\mathcal{I}, \sigma[y := d, x := e] \models p(x, y).$$

Define a function $h : D^{\mathcal{I}} \rightarrow D^{\mathcal{I}}$ by choosing, for each $d \in D^{\mathcal{I}}$, one such witness e and setting $h(d) := e$. Now define a new structure $\mathcal{I}'$ that extends $\mathcal{I}$ by interpreting the new function symbol g as h , i.e. $g^{\mathcal{I}'} = h$, and otherwise agrees with $\mathcal{I}$ on all symbols of the original language.

We claim that $\mathcal{I}' \models D$. Let $d \in D^{\mathcal{I}'}$ be arbitrary. By definition of h and $g^{\mathcal{I}'}$, we have $g^{\mathcal{I}'}(d) = h(d)$, and $h(d)$ is chosen so that

$$\mathcal{I}, \sigma[y := d, x := h(d)] \models p(x, y).$$

Since $\mathcal{I}'$ and $\mathcal{I}$ agree on the interpretation of p , it follows that

$$\mathcal{I}', \sigma[y := d] \models p(g(y), y).$$

Because d was arbitrary, we conclude that

$$\mathcal{I}' \models \forall y p(g(y), y),$$

so $\mathcal{I}' \models D$, and therefore D is satisfiable.

(D satisfiable $\Rightarrow C$ satisfiable). Conversely, suppose that D is satisfiable. Then there exists a structure $\mathcal{J}$ such that

$$\mathcal{J} \models \forall y p(g(y), y).$$

Let $h := g^{\mathcal{J}} : D^{\mathcal{J}} \rightarrow D^{\mathcal{J}}$ be the interpretation of g in $\mathcal{J}$.

We show that $\mathcal{J}$ is also a model of C when viewed as a structure for the original language (i.e. ignoring the extra symbol g in the statement of C). Let $d \in D^{\mathcal{J}}$ be arbitrary. Since

$$\mathcal{J} \models \forall y p(g(y), y),$$

we have, for every assignment σ ,

$$\mathcal{J}, \sigma[y := d] \models p(g(y), y).$$

Unfolding the term $g(y)$, this is the same as

$$\mathcal{J}, \sigma[y := d] \models p(h(d), y).$$

Equivalently,

$$\mathcal{J}, \sigma[y := d, x := h(d)] \models p(x, y).$$

Thus, for each $d \in D^{\mathcal{J}}$, there exists an element $e = h(d) \in D^{\mathcal{J}}$ such that

$$\mathcal{J}, \sigma[y := d, x := e] \models p(x, y).$$

By the semantics of $\exists$, this means

$$\mathcal{J}, \sigma[y := d] \models \exists x p(x, y)$$

for every $d \in D^{\mathcal{J}}$. By the semantics of $\forall$, we conclude that

$$\mathcal{J}, \sigma \models \forall y \exists x p(x, y),$$

i.e. $\mathcal{J}, \sigma \models C$. Hence C is satisfiable.

Since each direction holds, we conclude that C and D are equi-satisfiable. □